

A Stratified Reality Operator Derived from the Universal Quantifier

Formal semantics, deductive rules, limitations, and a proposed mathematical glyph

Formal proposal

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Abstract

This document proposes a new mathematical symbol, graphically derived from the universal quantifier $\forall$, but equipped with two horizontal bars. The symbol is intended to express the generic phrase “in reality” in a precise, stratified sense. Its lower stratum represents an imaginary, latent, or context-structuring level; its middle stratum represents energy, transformation, or structured development; and its upper stratum represents matter, observable results, certified consequences, or work that the development makes necessary. The two bars are interpreted as transition interfaces between these three levels.

The proposal is formalized as a non-vacuous universal operator over admissible realization chains. The operator is therefore not identified with the ordinary universal quantifier, nor with a standard modal actuality operator. It combines (i) existence of at least one admissible realization and (ii) invariance of the asserted proposition over every admissible realization chain. Introduction and elimination rules, closure properties, restriction principles, counterexamples, a distinction between observed manifestations and actionable obligations, and an optional interface with four-status certification systems are given. The document treats the glyph and its semantics as a new proposal, not as an already standardized notation.

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1 Purpose and formal status of the proposal

The ordinary universal quantifier $\forall$ binds a variable and asserts that a formula holds for every element of a specified domain. The expression “in reality” ordinarily behaves differently: it qualifies a proposition relative to a context, a realization process, and a manifested outcome. Consequently, the proposed symbol should not be introduced merely as a typographical variant of $\forall$. Its correct formal role is that of a *stratified actuality or realization operator* whose semantics is defined by a hierarchy of admissible realization chains.

The proposed glyph is denoted by

$$\forall.$$

It is visually an inverted letter *A*, like $\forall$, but with two horizontal bars. These bars partition the sign into three conceptual strata. The intended generic reading is

$$\forall\varphi \quad \text{“in reality, } \varphi\text{”}.$$

Whenever ambiguity is possible, the hierarchy is displayed explicitly:

$$\forall_I[E]^M \varphi.$$

The lower index *I* names the structural context, the bracketed term *E* names the structured development, and the upper index *M* names the manifestation layer.

Convention 1.1 (Provisional names). The operator may be called the *stratified reality operator*, the *stratified actuality operator*, or the *non-vacuous hierarchical universal operator*. This document uses “stratified reality operator” and writes its semantic counterpart as $R_{\mathfrak{S}}$.

Remark 1.2 (No standardization claim). The glyph introduced here is a private mathematical symbol generated in L^AT_EX. No claim is made that it is an existing Unicode character or an established symbol of mathematical logic.

2 Graphic semantics of the three strata

The sign is read from bottom to top:

$$\begin{array}{c} M : \text{manifestation, matter, observable consequence, or identified work,} \\ \hline E : \text{energy, transformation, calculation, or structured development,} \\ \hline I : \text{imaginary, latent, contextual, or structure-generating level.} \end{array}$$

The adjective *imaginary* is not used here as a synonym for false or fictitious. It refers to a non-manifest level that can organize, constrain, or generate the development. Depending on

the application, I may contain assumptions, boundary conditions, a geometric architecture, a formal grammar, an ambient category, a model class, or a latent state.

The middle level E contains the effective transformation of the context: a proof, a computation, a dynamical evolution, an algorithm, a chain of constructions, or an energy-carrying process. The upper level M contains what becomes manifest, verifiable, actionable, or materially instantiated.

The two horizontal bars represent two distinct transition interfaces:

$$I \xrightarrow{\alpha} E \xrightarrow{\mu} M.$$

The lower interface α is an *activation interface*: it asks whether the structural context genuinely supports the development. The upper interface μ is a *manifestation interface*: it asks whether the development genuinely produces, certifies, or requires the upper-level object.

3 Hierarchical realization frames

Definition 3.1 (Stratified realization frame). *A stratified realization frame is a tuple*

$$\mathfrak{H} = (I, E, M, A_{IE}, A_{EM}),$$

where:

- (i) I is a set of structural contexts;
- (ii) for every $i \in I$, $E(i)$ is a set of candidate structured developments;
- (iii) for every $i \in I$ and $e \in E(i)$, $M(i, e)$ is a set of candidate manifestations;
- (iv) $A_{IE}(i, e)$ is the admissibility predicate for the transition from i to e ;
- (v) $A_{EM}(i, e, m)$ is the admissibility predicate for the transition from (i, e) to m .

The symbols E and M in Definition 3.1 are dependent families. This is essential: the admissible developments may depend on the chosen context, and the admissible manifestations may depend on both the context and the development.

Definition 3.2 (Admissible realization chain). *An admissible realization chain of $\mathfrak{H}$ is a triple*

$$c = (i, e, m)$$

such that

$$i \in I, \quad e \in E(i), \quad m \in M(i, e),$$

and

$$A_{IE}(i, e), \quad A_{EM}(i, e, m).$$

The set of all admissible realization chains is

$$\text{Ch}(\mathfrak{H}) = \left\{ (i, e, m) \left| \begin{array}{l} i \in I, \ e \in E(i), \ m \in M(i, e), \\ A_{IE}(i, e), \ A_{EM}(i, e, m) \end{array} \right. \right\}.$$

Definition 3.3 (Realizability). *The frame $\mathfrak{H}$ is realizable if and only if*

$$\text{Ch}(\mathfrak{H}) \neq \emptyset.$$

Realizability is a constitutive part of the meaning “in reality”. A hierarchy with no admissible chain does not establish any proposition by vacuity.

Example 3.4 (Typical admissibility predicates). In a proof-theoretic application one may set

$$A_{IE}(i, e) := \text{Consistent}(i) \wedge \text{Licensed}(e \mid i) \wedge \text{Stable}(e),$$

and

$$A_{EM}(i, e, m) := \text{Traceable}(m, e) \wedge \text{Checkable}(m) \wedge \text{Reproducible}(m).$$

The exact predicates are application-dependent; the abstract logic below does not presuppose these particular choices.

4 Semantics of the stratified reality operator

Let formulas be interpreted relative to admissible chains. We write

$$\mathfrak{H}, c \models \varphi$$

for “ φ holds at chain c in frame $\mathfrak{H}$ ”.

Definition 4.1 (Stratified reality operator). *For a stratified realization frame $\mathfrak{H}$, define*

$$\mathfrak{H} \models \mathbf{R}_{\mathfrak{H}}\varphi$$

if and only if

$$\text{Ch}(\mathfrak{H}) \neq \emptyset \quad \text{and} \quad \forall c \in \text{Ch}(\mathfrak{H}), \mathfrak{H}, c \models \varphi.$$

In the proposed notation,

$$\mathbf{R}_{\mathfrak{H}}\varphi \equiv \forall_I[E]^M \varphi.$$

Thus the operator combines two semantic clauses:

- (a) *realization existence*: at least one admissible context–development–manifestation chain exists;
- (b) *realization invariance*: the formula holds on every admissible chain.

It is useful to separate these clauses meta-logically. Define

$$\text{Real}(\mathfrak{H}) :\Longleftrightarrow \text{Ch}(\mathfrak{H}) \neq \emptyset$$

and

$$\Box_{\mathfrak{H}}\varphi :\Longleftrightarrow \forall c \in \text{Ch}(\mathfrak{H}), \mathfrak{H}, c \models \varphi.$$

Then

$$\mathbf{R}_{\mathfrak{H}}\varphi \Longleftrightarrow \text{Real}(\mathfrak{H}) \wedge \Box_{\mathfrak{H}}\varphi.$$

The analogy with modal necessity is limited: $\Box_{\mathfrak{H}}$ quantifies over admissible realization chains, while $\text{Real}(\mathfrak{H})$ explicitly prevents vacuous truth.

Definition 4.2 (Contextual reading). *The formula*

$$\forall_I[E]^M \varphi$$

is read:

Relative to the structural contexts I , through the admissible structured developments E , and at the level of the admissible manifestations M , there exists at least one realization chain and φ holds on every such chain.

Its concise reading is “in reality, φ ”.

5 Basic deductive rules

Proposition 5.1 (Introduction rule). *If*

$$\text{Ch}(\mathfrak{H}) \neq \emptyset$$

and

$$\forall c \in \text{Ch}(\mathfrak{H}), \mathfrak{H}, c \models \varphi,$$

then

$$\mathfrak{H} \models R_{\mathfrak{H}}\varphi.$$

Proof. The two assumptions are exactly the two conjuncts in Definition 4.1. Therefore $\mathfrak{H} \models R_{\mathfrak{H}}\varphi$. $\square$

Proposition 5.2 (Elimination rule). *If*

$$\mathfrak{H} \models R_{\mathfrak{H}}\varphi$$

and $c \in \text{Ch}(\mathfrak{H})$, then

$$\mathfrak{H}, c \models \varphi.$$

Proof. By Definition 4.1, φ holds at every member of $\text{Ch}(\mathfrak{H})$. Since $c \in \text{Ch}(\mathfrak{H})$, it holds at c . $\square$

Proposition 5.3 (Non-vacuity). *If $\text{Ch}(\mathfrak{H}) = \emptyset$, then*

$$\mathfrak{H} \not\models R_{\mathfrak{H}}\varphi$$

for every formula φ .

Proof. Definition 4.1 requires $\text{Ch}(\mathfrak{H}) \neq \emptyset$. Hence the operator is false whenever the chain set is empty, independently of φ . $\square$

Corollary 5.4 (Truth and falsity constants). *For every frame $\mathfrak{H}$,*

$$\mathfrak{H} \models R_{\mathfrak{H}}\top \iff \text{Ch}(\mathfrak{H}) \neq \emptyset,$$

and

$$\mathfrak{H} \not\models R_{\mathfrak{H}}\perp.$$

Proof. The formula $\top$ holds at every chain, so Definition 4.1 reduces to the non-emptiness of $\text{Ch}(\mathfrak{H})$. For $\perp$, if the chain set is empty, Proposition 5.3 applies; if it is nonempty, $\perp$ fails on every chain. Hence $R_{\mathfrak{H}}\perp$ never holds. $\square$

6 Closure properties

Theorem 6.1 (Distribution over conjunction). *For all formulas φ and ψ ,*

$$R_{\mathfrak{H}}(\varphi \wedge \psi) \iff R_{\mathfrak{H}}\varphi \wedge R_{\mathfrak{H}}\psi.$$

Proof. Assume $R_{\mathfrak{H}}(\varphi \wedge \psi)$. Then $\text{Ch}(\mathfrak{H}) \neq \emptyset$, and for every $c \in \text{Ch}(\mathfrak{H})$ both φ and ψ hold at c . Therefore $R_{\mathfrak{H}}\varphi$ and $R_{\mathfrak{H}}\psi$.

Conversely, assume $R_{\mathfrak{H}}\varphi$ and $R_{\mathfrak{H}}\psi$. Each assumption implies $\text{Ch}(\mathfrak{H}) \neq \emptyset$. For every $c \in \text{Ch}(\mathfrak{H})$, both formulas hold, hence $\varphi \wedge \psi$ holds. Therefore $R_{\mathfrak{H}}(\varphi \wedge \psi)$. $\square$

Theorem 6.2 (Closure under modus ponens). *If*

$$R_{\mathfrak{H}}(\varphi \rightarrow \psi) \quad \text{and} \quad R_{\mathfrak{H}}\varphi,$$

then

$$R_{\mathfrak{H}}\psi.$$

Proof. Both assumptions imply $\text{Ch}(\mathfrak{H}) \neq \emptyset$. Let $c \in \text{Ch}(\mathfrak{H})$. By elimination, $\varphi \rightarrow \psi$ holds at c , and φ holds at c . By ordinary modus ponens at the chain level, ψ holds at c . Since c was arbitrary, ψ holds on every admissible chain, and therefore $R_{\mathfrak{H}}\psi$. $\square$

Theorem 6.3 (Monotonicity under semantic consequence). *Suppose that for every $c \in \text{Ch}(\mathfrak{H})$,*

$$\mathfrak{H}, c \models \varphi \implies \mathfrak{H}, c \models \psi.$$

Then

$$R_{\mathfrak{H}}\varphi \implies R_{\mathfrak{H}}\psi.$$

Proof. Assume $R_{\mathfrak{H}}\varphi$. Then $\text{Ch}(\mathfrak{H}) \neq \emptyset$, and φ holds at every chain. By the assumed chainwise consequence, ψ also holds at every chain. The non-emptiness condition is unchanged, so $R_{\mathfrak{H}}\psi$. $\square$

7 Restriction and hierarchy refinement

Definition 7.1 (Subhierarchy). *A frame $\mathfrak{K}$ is a subhierarchy of $\mathfrak{H}$, written $\mathfrak{K} \preccurlyeq \mathfrak{H}$, if*

$$\text{Ch}(\mathfrak{K}) \subseteq \text{Ch}(\mathfrak{H}).$$

It is a realizable subhierarchy if also $\text{Ch}(\mathfrak{K}) \neq \emptyset$.

Theorem 7.2 (Restriction principle). *Let $\mathfrak{K} \preccurlyeq \mathfrak{H}$ be a realizable subhierarchy. Then*

$$R_{\mathfrak{H}}\varphi \implies R_{\mathfrak{K}}\varphi.$$

Proof. Assume $R_{\mathfrak{H}}\varphi$. Then φ holds at every chain in $\text{Ch}(\mathfrak{H})$. Since $\text{Ch}(\mathfrak{K}) \subseteq \text{Ch}(\mathfrak{H})$, it holds at every chain in $\text{Ch}(\mathfrak{K})$. By hypothesis $\text{Ch}(\mathfrak{K}) \neq \emptyset$. Hence $R_{\mathfrak{K}}\varphi$. $\square$

Remark 7.3 (Failure of the converse). The converse of Theorem 7.2 is generally false. A formula can hold on a restricted family of realizations and fail on an admissible chain omitted by that restriction.

Example 7.4 (Counterexample to the converse). Let

$$\text{Ch}(\mathfrak{H}) = \{c_1, c_2\}, \quad \text{Ch}(\mathfrak{K}) = \{c_1\},$$

and suppose φ is true at c_1 and false at c_2 . Then $R_{\mathfrak{K}}\varphi$ holds, while $R_{\mathfrak{H}}\varphi$ does not.

8 Disjunction, negation, and logical limitations

The stratified reality operator is universal over admissible chains. It therefore has the same non-distributive behavior over disjunction as an ordinary universal quantifier, with the additional non-vacuity clause.

Proposition 8.1 (One-way disjunction principle).

$$R_{\mathfrak{H}}\varphi \vee R_{\mathfrak{H}}\psi \implies R_{\mathfrak{H}}(\varphi \vee \psi).$$

The converse is not valid in general.

Proof. If $R_{\mathfrak{H}}\varphi$, then the chain set is nonempty and φ holds on every chain; therefore $\varphi \vee \psi$ holds on every chain. The same argument applies if $R_{\mathfrak{H}}\psi$. Thus the implication holds.

For failure of the converse, take two chains c_1, c_2 . Let φ be true only at c_1 and let ψ be true only at c_2 . Then $\varphi \vee \psi$ holds at both chains, so $R_{\mathfrak{H}}(\varphi \vee \psi)$ holds. However neither φ nor ψ holds on every chain, so neither $R_{\mathfrak{H}}\varphi$ nor $R_{\mathfrak{H}}\psi$ holds. $\square$

Proposition 8.2 (Negation asymmetry). *Under bivalent chain semantics,*

$$R_{\mathfrak{H}}\neg\varphi \implies \neg R_{\mathfrak{H}}\varphi.$$

The converse is not valid in general.

Proof. If $R_{\mathfrak{H}}\neg\varphi$, then the chain set is nonempty and φ is false at every chain. Hence φ cannot be true at every chain, so $R_{\mathfrak{H}}\varphi$ fails.

For failure of the converse, take two chains c_1, c_2 and let φ be true at c_1 and false at c_2 . Then $R_{\mathfrak{H}}\varphi$ fails, but $\neg\varphi$ is not true at every chain, so $R_{\mathfrak{H}}\neg\varphi$ also fails. $\square$

9 Relation to the ordinary universal quantifier

The new operator is derived graphically from $\forall$, but its semantics is not reducible to variable quantification alone. Nevertheless, precise reduction results are available.

Definition 9.1 (Chain-invariant formula). *A formula φ is chain-invariant in $\mathfrak{H}$ if for all $c, d \in \text{Ch}(\mathfrak{H})$,*

$$\mathfrak{H}, c \models \varphi \iff \mathfrak{H}, d \models \varphi.$$

Theorem 9.2 (Reduction for chain-invariant formulas). *Assume $\text{Ch}(\mathfrak{H}) \neq \emptyset$ and φ is chain-invariant in $\mathfrak{H}$. Then, for any $c_0 \in \text{Ch}(\mathfrak{H})$,*

$$R_{\mathfrak{H}}\varphi \iff \mathfrak{H}, c_0 \models \varphi.$$

Proof. If $R_{\mathfrak{H}}\varphi$, elimination gives $\mathfrak{H}, c_0 \models \varphi$.

Conversely, suppose $\mathfrak{H}, c_0 \models \varphi$. By chain invariance, φ holds at every $c \in \text{Ch}(\mathfrak{H})$. The chain set is nonempty by hypothesis, so $R_{\mathfrak{H}}\varphi$. $\square$

Corollary 9.3 (Trivial hierarchy). *If $\text{Ch}(\mathfrak{H}) = \{c_0\}$, then*

$$R_{\mathfrak{H}}\varphi \iff \mathfrak{H}, c_0 \models \varphi.$$

Proof. Every formula is chain-invariant on a singleton chain set, so Theorem 9.2 applies. $\square$

Definition 9.4 (Bound-variable extension). *Let $D(c)$ be a domain attached to each admissible chain. Define*

$$\forall_{\mathfrak{H}} x \in D \varphi(x)$$

by

$$\text{Ch}(\mathfrak{H}) \neq \emptyset \quad \text{and} \quad \forall c \in \text{Ch}(\mathfrak{H}), \forall x \in D(c), \mathfrak{H}, c \models \varphi(x).$$

This extension makes explicit why the symbol remains related to universal quantification: it can universally quantify both over realization chains and over objects internal to each chain. The outer non-vacuity condition still concerns realization chains. If non-vacuity of each $D(c)$ is also required by the application, that requirement must be stated separately.

10 Observed manifestations and actionable obligations

The upper stratum should distinguish what has been established from what has merely become necessary to do.

Definition 10.1 (Observed and actionable manifestations). *Suppose every manifestation family decomposes as a disjoint union*

$$M(i, e) = M^{\text{obs}}(i, e) \sqcup M^{\text{act}}(i, e).$$

The observed part M^{obs} contains results, objects, or consequences already manifested and checkable. The actionable part M^{act} contains tasks, proof obligations, measurements, constructions, or decisions that the development makes necessary.

Let $\mathfrak{H}^{\text{obs}}$ and $\mathfrak{H}^{\text{act}}$ be the corresponding restricted frames. Then

$$\forall_I[E]^{M^{\text{obs}}} \varphi$$

means that φ holds throughout the nonempty class of observed realization chains, whereas

$$\forall_I[E]^{M^{\text{act}}} \text{Obligation}(\varphi)$$

means that the obligation concerning φ is itself invariantly generated by every actionable realization chain.

Proposition 10.2 (No collapse of obligation into achievement). *Without an additional completion axiom,*

$$\forall_I[E]^{M^{\text{act}}} \text{Obligation}(\varphi) \not\Rightarrow \forall_I[E]^{M^{\text{obs}}} \varphi.$$

Proof. An actionable chain may identify a task without containing any observed chain in which the task has been completed. Formally, choose a frame with a nonempty actionable chain set and an empty observed chain set. The left-hand formula can hold, while the right-hand formula fails by non-vacuity. Therefore the implication is not valid without an axiom that maps completed obligations to observed manifestations. $\square$

Definition 10.3 (Completion bridge). *A completion bridge is a relation*

$$B \subseteq \text{Ch}(\mathfrak{H}^{\text{act}}) \times \text{Ch}(\mathfrak{H}^{\text{obs}})$$

such that $B(a, o)$ means that the actionable chain a has been completed and certified by the observed chain o .

A project-specific completion theorem may then establish an implication from an actionable reality statement to an observed reality statement, but such an implication is not part of the bare logic of $\forall$.

11 A certification layer and four-status compatibility

The semantics above is classically bivalent at the meta-level. In applications, one may need a separate certification status recording whether the chain set, the transition interfaces, and the universal check have been completely verified.

Definition 11.1 (Abstract certification status). A four-status interface is a map

$$\text{Stat}_{\mathfrak{H}}(\varphi) \in \{\text{F}, \text{NF}, \text{NT}, \text{T}\}$$

subject at least to the following soundness requirements:

$$\begin{aligned} \text{Stat}_{\mathfrak{H}}(\varphi) = \text{T} &\implies \text{R}_{\mathfrak{H}}\varphi, \\ \text{Stat}_{\mathfrak{H}}(\varphi) = \text{F} &\implies \exists c \in \text{Ch}(\mathfrak{H}) \text{ such that } \mathfrak{H}, c \not\models \varphi. \end{aligned}$$

The statuses NF and NT are intentionally left open for a project-specific HT+NT algebra; they do not, by themselves, entail classical truth or classical falsity.

Remark 11.2 (One compatible operational reading). A non-normative operational interpretation is:

- T: realizability and universal satisfaction are fully certified;
- F: a certified admissible counterexample exists;
- NT: no certified counterexample is known, but at least one admissible chain remains unresolved;
- NF: realization or interface certification is presently insufficient to form a certified nonempty chain family.

This reading is only an interface example. A pre-existing HT+NT calculus may assign more specific meanings, provided the two soundness implications above are preserved.

If a robust conjunction $\wedge_{\text{rob}}$ is supplied by the external status algebra, Theorem 6.1 can be mirrored at the certification level only after proving that the certification procedure for $\varphi \wedge \psi$ is equivalent to the robust combination of the separate certification procedures. This is an additional theorem about the certification system, not an automatic consequence of the classical chain semantics.

12 Examples

Example 12.1 (Proof program). Let

- $I = \mathcal{C}$ be a formal corpus of definitions, assumptions, and certified prior results;
- $E = \mathcal{D}$ be a family of admissible proof developments;
- $M = \mathcal{O}$ be the resulting certificates, counterexamples, or proof obligations.

Then

$$\forall_{\mathcal{C}}[\mathcal{D}]^{\mathcal{O}^{\text{obs}}} \text{Stable}(\Phi)$$

asserts that at least one admissible certified development exists and that every observed realization chain certifies the stability of Φ .

By contrast,

$$\forall_{\mathcal{C}}[\mathcal{D}]^{\mathcal{O}^{\text{act}}} \text{Obligation}(\text{Prove}(\Phi))$$

asserts only that the need to prove Φ is an invariantly generated task. It does not assert Φ .

Example 12.2 (Numerical computation). Let I encode input assumptions and precision constraints, E encode certified algorithms and execution traces, and M encode numerical outputs with error bounds. Then

$$\forall_I[E]^M |\hat{x} - x| \leq \varepsilon$$

means that there is at least one admissible certified computation and that every admissible computation chain produces an approximation satisfying the required bound.

Example 12.3 (Physical model). Let I be a family of admissible latent states and boundary conditions, E a family of evolutions, and M a family of measurements. The statement

$$\forall_I[E]^{M^{\text{obs}}} \varphi$$

should not be read as metaphysical reality without qualification. It means that φ is invariant over the explicitly selected and nonempty family of admissible model-to-measurement chains. The frame must therefore be declared whenever the application is scientifically consequential.

13 Design constraints for rigorous use

A rigorous use of the symbol should satisfy the following constraints.

- D1. Frame declaration.** The context family I , development family E , manifestation family M , and both admissibility predicates must be declared or referenced.
- D2. Non-vacuity.** A proof of $\text{Ch}(\mathfrak{H}) \neq \emptyset$ is part of every introduction of $\forall$.
- D3. No hidden change of hierarchy.** Replacing $\mathfrak{H}$ by a subhierarchy or superhierarchy must be stated explicitly.
- D4. Observed/actionable distinction.** A proof obligation must not be written as an observed theorem.
- D5. Traceability.** The two bars should correspond to checkable transition predicates, not merely to an informal metaphor.
- D6. No ontological overclaim.** The operator asserts reality relative to a declared realization frame. It does not, by notation alone, settle an unrestricted metaphysical notion of reality.
- D7. Certification separation.** Logical truth on admissible chains and the practical status of having verified all chains are distinct notions.

14 Recommended formal definition

The most stable compact definition is the following:

$$\boxed{\forall_I[E]^M \varphi \quad :\Longleftrightarrow \quad \text{Ch}(I, E, M) \neq \emptyset \wedge \forall (i, e, m) \in \text{Ch}(I, E, M), \varphi(i, e, m)}$$

where

$$\text{Ch}(I, E, M) = \left\{ (i, e, m) \left| \begin{array}{l} i \in I, \ e \in E(i), \ m \in M(i, e), \\ A_{IE}(i, e), \ A_{EM}(i, e, m) \end{array} \right. \right\}.$$

In words:

“In reality” means that at least one admissible realization exists and that the asserted proposition is invariant over all admissible realizations of the declared context–development–manifestation hierarchy.

15 Conclusion

The proposed symbol preserves the universal visual intuition of $\forall$ while adding two semantically meaningful interfaces. Its lower, middle, and upper strata encode context, development, and manifestation. Formally, its decisive feature is not the three-part picture alone, but the conjunction of non-vacuous realizability and universal validity over admissible chains.

The operator is therefore stronger than a bare contextual phrase and different from ordinary universal quantification. It can express observed reality, generated obligations, certified numerical invariance, or model-relative manifestation, provided that the underlying frame and admissibility predicates are explicitly declared. Its use is mathematically legitimate as a newly defined operator, while its broader adoption would require stable notation, examples, and compatibility conventions across domains.

A Reusable L^AT_EX definition of the glyph

The following code is self-contained once `tikz` is loaded:

```
\newcommand{\realforallglyph}[1]{%
  \tikz[
    baseline=-0.15em,
    x=0.75em,
    y=0.78em,
    scale=#1,
    line cap=round,
    line join=round
  ]{%
    \draw[line width=.036em]
      (-0.44,0.96) -- (0,0) -- (0.44,0.96);
    \draw[line width=.034em]
      (-0.31,0.68) -- (0.31,0.68);
    \draw[line width=.034em]
      (-0.19,0.41) -- (0.19,0.41);
  }%
}
```

```
\DeclareRobustCommand{\Rforall}{%
  \mathord{%
    \mathchoice
      {\realforallglyph{1.00}}
      {\realforallglyph{0.90}}
      {\realforallglyph{0.70}}
      {\realforallglyph{0.55}}}%
}
```

Example:

$$\forall_I[E]^{M^{\text{obs}}} \varphi.$$

B Selected background references

References

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